Chapter route

Which cryptographic and system mechanisms provide verifiable guarantees when identities, programs, and networks can be attacked?

Prerequisites are protection, basic networking, probability, and hashing.

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Learning objectives

- Distinguish confidentiality, integrity, availability, and authentication.
- Analyze symmetric/asymmetric encryption and hashes.
- Evaluate defenses under explicit attack models.

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The security problem

Specify the threat model and CIA goals. Identity, authorization, and freshness differ. Valid messages still require receiver state against replay.

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Assets, adversaries, and goals

Confidentiality restricts unauthorized reading; integrity restricts unauthorized modification; availability requires authorized service under stated conditions. A threat can violate a goal, a vulnerability is an exploitable weakness, and an attack is an actual attempt. Theft followed by encryption/deletion can violate several goals.

The teaching grade service protects grades, credentials, and availability. The adversary controls the network and submits arbitrary forms, but cannot modify the kernel or obtain the server's private key. If key theft is later allowed, earlier authentication conclusions no longer follow. Every guarantee depends on its adversary, trust roots, keys, and endpoint integrity.

Physical, human, OS, and network factors affect the boundary. TLS does not protect input captured by an endpoint keylogger. ACLs restrict unauthorized use without proving every valid-account action follows business rules.

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Impersonation, replay, and intermediaries

Masquerading impersonates an identity; replay resends old valid messages; a man-in-the-middle controls what both endpoints observe. Source IP is not cryptographic identity, and receiving a public key does not bind it to a service.

A valid MAC on transfer(A, B,100, id17) can be replayed even without understanding its content. Verification proves a key holder produced that content, not fresh authorization. Authenticate session, sequence, nonce, and operation ID, and maintain receiver usage state. Returning an earlier result on retries requires durable deduplication atomically linked to the transaction.

Distinguish inability to forge a new valid operation from inability to execute one authorization repeatedly. Encryption alone establishes neither combined claim.

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Program threats

Separate execution authority from infection, triggering, and concealment. Memory-safety claims need explicit byte bounds.

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Authority and propagation

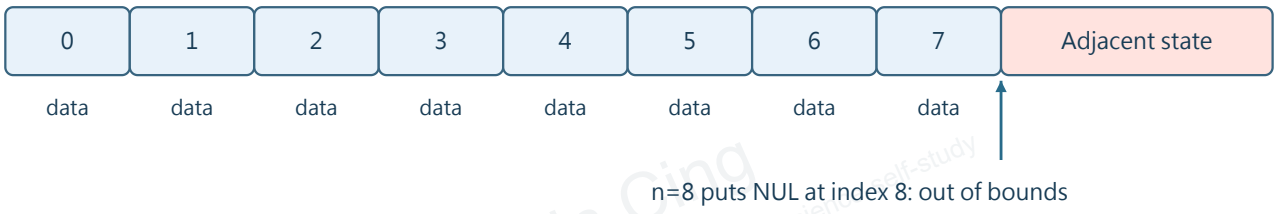
A Trojan abuses execution authority behind apparent useful behavior. A backdoor bypasses normal entry; a logic bomb triggers under a condition. Viruses infect host files/programs, while worms propagate through services/networks. Categories overlap and do not restrict viruses to one OS vendor.

An untrusted directory in PATH can substitute a namesake tool. Malicious code acquires the victim program's actual rights, not automatic root authority from its label. A compromised compiler can produce malicious output from inspected source. Executables, loaders, compilers, and update channels belong in the trust boundary.

File, boot, macro, and source viruses describe infection locations. Polymorphic, encrypted, stealth, and armored describe evasion; multipartite affects several locations. Byte signatures and behavior analysis complement one another. No signature match does not prove absence.

Buffer bounds precede control-flow claims

Abstract byte layout: the string terminator occupies a byte



Capacity 8 holds at most 7 data bytes + NUL. Binary buffers use explicit lengths.

An eight-byte buffer has indices 0–7. A C string with n data bytes requires $n+1 \leq 8$ for complete copying. At n=8 the data fits, but its NUL at index 8 overflows. Actual layout depends on ABI/compiler; this abstract figure illustrates adjacent corruption without platform addresses.

Buffer bounds precede control-flow claims (continued)

One safe contract rejects overlength input. Another explicitly truncates to k=min(n,7), copies k bytes, and sets buf[k]=0. Strncpy(...,7) alone need not append NUL. Binary data requires explicit lengths rather than strlen, with range and integer-overflow checks.

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Different defenses prove different properties

Bounds/type/lifetime checks address invalid accesses. Canaries detect some overwrites before return; NX/W⊕X restrict execution from data pages; ASLR complicates address prediction; control-flow restrictions constrain targets. NX neither prevents out-of-bounds writes nor alone excludes reuse of existing executable code.

Corrupting is_admin, length, or an object pointer can violate authorization without changing return addresses. Least privilege limits a damaged component's reach but cannot replace fixing the overflow. A managed safe subset eliminates some memory errors while native/unsafe code, runtime flaws, and business policy remain boundaries. Type safety does not prove every security goal.

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System and network threats

Model exposure, propagation, and denial of service through resources/state. Pending allocation rate and retention time jointly determine occupancy.

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Worm propagation and service exposure

The historical 1988 Morris worm exploited service weaknesses and remote trust, then sought reachable hosts. Repeated infection consumed resources, demonstrating that a security event need not steal data.

In the original discrete model, each infected host infects at most two fresh hosts per round, without duplication or removal. $I_0=1$ and $I_{t+1}=3I_t$ give 1,3,9,27. This ideal early upper-bound model is not Morris measurement. A closed 100-host network instead satisfies $I_{t+1} \leq \min(100,3I_t)$; it cannot contain 243 infected hosts.

Scanning observes reachable services. An open port does not prove a vulnerability, and banners need not be truthful. Authorization, patching, and disabling unnecessary services reduce exposure; removing one entry does not fix input bugs in remaining services.

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Resource exhaustion and defensive side effects

A service has 1000 pending slots retained 10 s per incomplete handshake. Without early cleanup, 100 arrivals/s fills 1000 on average under L=λW. Attackers need not log in if they can force scarce-state allocation. Cookies, quotas, and delayed allocation change that model without creating infinite upstream bandwidth.

Legitimate 80/s plus attack 40/s exceeds 100/s service by at least 20/s. Rate limits must name their key, such as user, IP, tenant, or global quota. IP limits can penalize users sharing NAT. Five wrong passwords causing a ten-minute lock blocks guessing but lets attackers lock others out. DDoS uses coordinated sources; a flash crowd is legitimate excess demand. High load alone does not identify malice. Responses trade availability against false blocking.

Cryptography

Each primitive supplies a specific guarantee; identity binding and freshness require further arguments. RSA correctness, nonce reuse, and historical TLS require explicit derivations.

Primitives and key roles

Tool	Confidentiality	Integrity/source	Creation and verification
Symmetric encryption	With a suitable scheme	Not from encryption alone	Shared-secret holders
Public-key encryption	With a suitable scheme	Does not establish sender	Anyone encrypts; private key decrypts
Hash	No	Mismatch against trusted digest	Anyone computes
MAC	No	Shared-key origin/integrity	Key holders generate/verify
Signature	No	Private-key origin/integrity	Private signs; public verifies

Correctness means $D_k(E_k(m))=m$. Security is a computational claim against specified distinguishing/forgery attacks, not logical impossibility of guessing m. Algorithms can be public; secret keys remain secret. Public keys need authenticated integrity of their identity binding.

Reused keystream algebra

For $c=m \oplus s$, $m=c \oplus s$. Let m_1=0x41, m_2=0x42, s=0xA6; c_1=0xE7, c_2=0xE4. Reuse gives $c_1 \oplus c_2$=0x03=$m_1 \oplus m_2$. Knowing m_1 reveals s and then m_2. This identity does not break the underlying cipher.

Block ciphers process fixed blocks; longer messages need correct modes and nonce/IV rules. AES's 128-bit block differs from 128/192/256-bit keys. DES, 3DES, RC4, WEP are historical context, not new-deployment recommendations. Use a complete analyzed confidentiality/authentication scheme. An unprotected public hash appended to ciphertext can be recomputed after tampering.

A complete small RSA computation

This number-theory example is not secure encryption. Set p=5, q=11, n=55,ϕ=40, e=3, d=27, ed=81=1+2ϕ. Public key=(55,3), private=(55,27). For m=12, 12^2 mod55=34 and 12^3 mod55=34 • 12 mod55=23.

Decrypt with square table $23^1 \to 23, 23^2 \to 34, 23^4 \to 1, 23^8 \to 1, 23^{16} \to 1$. Since 27=16+8+2+1, 23^{27} mod55=1 • 1 • 34 • 23 mod55=12. Raw RSA is deterministic and algebraically structured. Deployable encryption/signatures additionally require specified encoding/padding, key sizes, and purpose separation. These small primes provide no confidentiality.

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RSA correctness for every message

For every m∈{0,…, n−1}, ed=1+k(p−1)(q−1). Modulo p, if p divides m, both m and m^ed are 0. Otherwise Fermat gives m^(p−1)≡1, so

$$m^{ed} = m(m^{p-1})^{k(q-1)} \equiv m \pmod{p}.$$

The same branches hold modulo q. Distinct primes and CRT yield m^ed≡m modpq, including messages not coprime with n.

Euler applied only when gcd(m, n)=1 would omit valid multiples of p/q. This proves mathematical decryption correctness, not RSA hardness, chosen-ciphertext security, or side-channel resistance.

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Hashes, MACs, and signatures

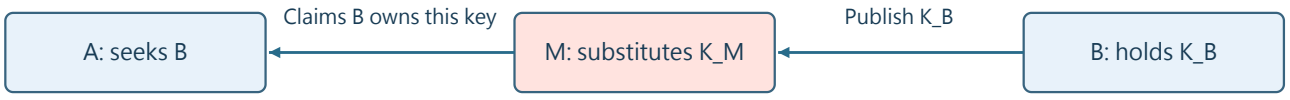
An n-bit hash has 2^n outputs, so collisions exist by the pigeonhole principle for a larger message space. Collision resistance concerns finding any distinct pair with equal digest; second-preimage resistance fixes one message first. Digest equality is computational evidence, not a theorem that messages are equal.

Anyone can replace (m, H(m)) with (m′, H(m′)); public hashes do not authenticate. MAC verifiers also know the secret, so MAC alone cannot prove to a third party that only the other participant generated a tag. Signatures use private signing and public verification, with trusted key binding, complete encoding, and key protection. They are not arbitrary encryption run backward.

HMAC is a standard keyed construction, not arbitrary H(key ∥ m). Include context, roles, session, and sequence against cross-purpose reuse.

Key distribution and substitution

Public keys need trusted identity binding, not secrecy



Without binding: A encrypts under K_M → M decrypts. Correct cipher, wrong identity.

Validate chain, name, purpose, validity, and handshake proof of private-key possession.

A seeking B receives an unauthenticated claim "B, key K". M substitutes K_M; A encrypts to M, which can decrypt although the cipher works correctly. Certificates bind service identity and key. Validate signature chains to trusted roots, target name, purpose, time, and relevant revocation policy. Certificate presence alone is insufficient.

Key distribution and substitution (continued)

Pairwise symmetric keys for N users require N(N−1)/2 keys. Six users need 15; adding a seventh needs six more. A KDC reduces long-term secrets per user but centralizes trust. Public-key techniques reduce secret distribution without eliminating public-key authentication or private-key protection.

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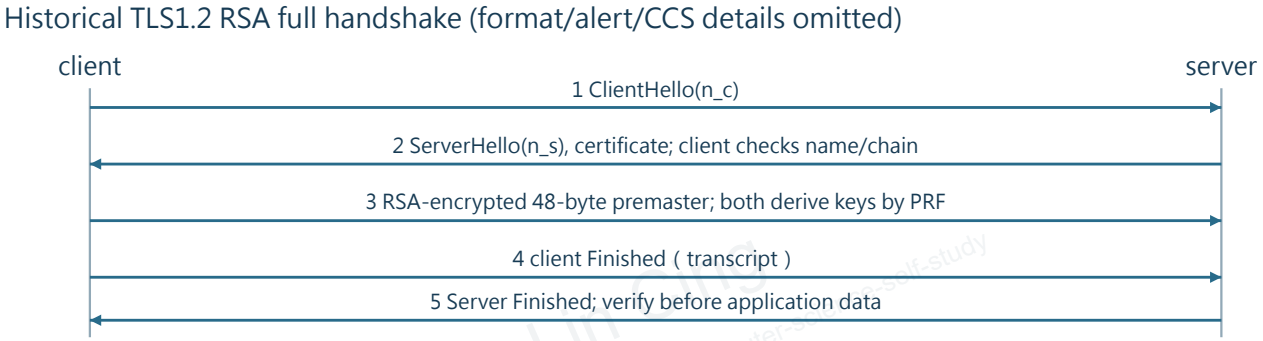
Endpoints determine protection scope

IPsec protects IP traffic between its network endpoints; TLS protects its session endpoints; encrypted mail can span store-and-forward hops. TLS termination at a reverse proxy exposes plaintext there and does not automatically protect proxy-to-backend traffic.

Link/network protection does not directly identify an application user. Disk encryption protects offline media, not a logged-in process authorized to read plaintext. For each layer state key holders, plaintext observers, and trusted components. Layer counts do not sum into unconditional safety.

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Historical TLS 1.2 RSA handshake



Use RFC5246 full handshake with RSA key exchange, not TLS1.3. ClientHello carries n_c; ServerHello carries n_s and certificate. Client validates chain, target name, purpose, and time, constructs a 48-byte premaster secret, two version bytes plus 46 random bytes, and sends its RSA-encoded encryption. Both sides use the PRF to derive master and directional traffic keys.

Historical TLS 1.2 RSA handshake (continued)

Both must verify Finished authentication of the prior handshake transcript. A certificate alone proves a CA bound that key, not current private-key possession or untampered handshake execution. With static RSA exchange, later private-key compromise and a recorded handshake can reveal old premaster secrets even after traffic keys were deleted. It lacks forward secrecy.

SSL3 and TLS are not synonyms. RFC7568 prohibited SSL3 in 2015. This historical example explicitly limits its version and omitted details.

Freshness and atomic nonce consumption

Server issues nonce 42 with pending[42]=(U, expiry). Client returns MAC_k("login" ∥ U ∥ n ∥ serverID). Receiver validates tag and pending/expiry, and atomically changes n to used. First succeeds; an identical valid replay fails because n is used.

MAC integrity cannot reject a copied tag without freshness state. Consume only after successful verification and make checking/consumption atomic; otherwise two threads can both observe pending. Random nonces need entropy/nonreuse; counters need restart persistence or a new session. Payments additionally link consumption and business effects transactionally or through recoverable deduplication, preventing crash-induced loss/repetition.

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User authentication

Passwords, OTPs, and biometrics use different error/cost denominators. Salt separates record computations without eliminating offline guesses.

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Credentials and sessions

Knowledge, possession, and inherence are distinct factor types. Two passwords remain one type; MFA uses at least two types, not necessarily all three. A stolen authenticated session token can bypass the original factors, so session protection, invalidation, and reauthentication matter.

An abstract verifier record is {user, salt, cost, KDF(password, salt, cost)}. Login recomputes and compares. Public salt prevents sharing one precomputed dictionary across records, not guessing one record or mathematically eliminating collisions. Stolen verifiers still permit offline guesses and need access protection. Historical eight-character UNIX limits belong to particular old schemes, not every UNIX or a modern policy.

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Guess spaces and computation

Uniform random four-digit decimal PINs, including 0000, have 10000 possibilities. Distinct unbiased guesses have mean rank 5000.5. At 1 ms/guess, mean time is 5.0005 s. This is not every four-character password or a human-biased dictionary.

For 1000 distinct salts and 100000 candidates, testing every record costs about $1000 \cdot 100000$ KDFs rather than 100000 reusable hashes. At 5 ms each, one serial work unit takes 500000 s$\approx$5.79 days. Parallel resources, memory constraints, and prior distributions change actual cost. Online limits do not constrain stolen-verifier offline work; lockouts can become DoS.

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OTP and biometric errors

OTP/challenge-response adds freshness. One-time use is a receiver acceptance rule, not merely changing displayed digits. TOTP windows need explicit replay/used-code handling and remain vulnerable to real-time phishing relay.

Biometric thresholds trade false acceptance and rejection. FAR uses impostor attempts; FRR uses genuine attempts. With 100000 impostor attempts, FAR 0.1% means 100 false accepts; with 10000 genuine attempts, FRR 2% means 200 false rejects. Threshold, population, device, and environment change estimates.

Biometrics are not freely replaceable secrets. Templates, liveness/attack detection, lost devices, and fallback policy affect security. Factors sharing a compromised endpoint/recovery path can fail together; multiplying failure probabilities requires independence.

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Implementing defenses

Connect controls to attack paths. Interpret IDS alarms with Bayes. Baseline, checker, and log integrity are trust assumptions.

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Policy, assessment, and layers

Policy specifies assets, operations, responsibilities, and response. Assessment checks configuration, code, and exposure; authorized penetration tests examine exploit paths. A passed scan supports its time/scope/tools/known checks, not universal absence of vulnerabilities.

Layers sharing a stolen key or bad identity header have common failure. Patching, least privilege, memory/input safety, segmentation, authentication, monitoring, and recovery affect different probabilities/damage. Name the path each blocks.

An original model has loss 100000 currency units and annual probability 0.02, expected loss 2000/year. Reducing probability to 0.005 leaves 500, saving expected 1500/year. Tail disasters, correlated events, and regulatory obligations are absent and can still determine policy.

IDS/IPS and base rates

Signatures detect encoded patterns; anomaly detectors find baseline deviations. Unknown attacks need not all evade signatures; anomalous events need not be malicious. IDS usually alerts; IPS can block, directly risking availability on errors.

Of 1000000 events, 100 are attacks. TPR=0.9 gives TP=90, FN=10. Normal=999900 with FPR=0.001 gives expected FP=999.9, TN=998900.1. Alarm precision=90/(90+999.9)≈8.26%, not 90%. TPR divides by attacks; precision divides by alarms.

$$\text{precision} = \frac{\pi TPR}{\pi TPR + (1-\pi)FPR}.$$

At π=0.0001, TPR=0.9, precision≥0.5 requires FPR≤0.000090009…. Baseline contamination, drift, and correlated-event aggregation affect estimation. Development data cannot be relabeled independent acceptance data.

Integrity monitoring and recovery

Compare content/metadata with a trusted baseline. Compromising both checker and baseline removes that trust root. Hash comparisons require suitable collision resistance and protected original digests; equality is not absolute identity proof.

Tripwire-style monitoring distinguishes authorized updates from unexpected changes. Changing logs still need protected remote/append-only records and correlation. A honeypot must not become an unrestricted internal attack relay. Antivirus signatures, behavior, and sandboxes have limited coverage; a clean scan does not prove supply-chain innocence.

Response includes isolation, evidence preservation, root-cause repair, trusted-backup recovery, and credential rotation. Audit subject, object, action, decision, time, and correlation ID while protecting log integrity/availability.

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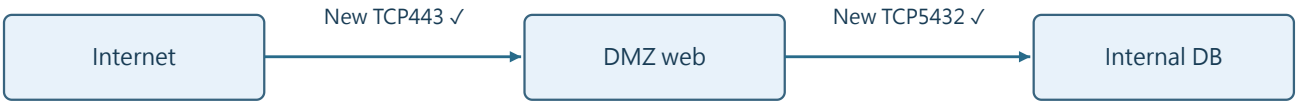
Firewalls

Specify zones, direction, port, state, and first-match rules. Vulnerabilities inside allowed services remain exploitable.

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Allowed connections before filter rules

Zones are not full trust: allow only specified directions/services



Deny new Internet → DB and web → DB22. Validate return packets against existing state.

Compromised web still reaches DB5432. DB needs authentication and minimum query authority.

Allow new Internet→DMZ web TCP443, web→internal DB TCP5432, and internal→Internet TCP443. Deny other new connections. Permit returns only if they match existing state. DMZ does not gain all internal access. A compromised web server can abuse its required DB5432 path, requiring database authentication/query authorization.

Allowed connections before filter rules (continued)

Stateless filters inspect headers; stateful filters track connections; proxies terminate/understand applications; host firewalls govern local ingress/egress. Without TLS termination, packet filters cannot inspect encrypted application meaning. Allowing 443 does not make its payload safe.

Rule order and limits

Rules2–4 implement the three new-connection paths: Internet→web, web→DB, and internal→Internet. Rule1 admits subsequent packets in either direction of an existing allowed connection. New TCP service rules match destination ports.

Order	Match	Action
1	Packet matching an existing allowed connection state, either direction	Allow
2	New TCP, Internet→DMZ web, destination443	Allow
3	New TCP, DMZ web→internal DB, destination5432	Allow
4	New TCP, internal→Internet, destination443	Allow
5	All remaining packets	Deny

Rule order and limits (continued)

Packet	First matching rule	Result
New Internet→web TCP443	2	Allow
New web→DB TCP5432	3	Allow
New internal→Internet TCP443	4	Allow
New Internet→DB TCP5432	5	Deny
New web→DB TCP22	5	Deny
DB→web return actually matching existing state	1	Allow
web→DB subsequent data in an existing allowed connection	1	Allow

Match source/destination addresses, zones, destination port, direction, and state. An ACK bit or a claimed established status without a matching state-table entry does not satisfy rule1.

Rule order and limits (continued)

Putting deny-all first makes later permits unreachable. Actual products have specific chain/priority semantics. Firewalls do not repair allowed-protocol bugs, authenticate hosts merely by IP, or restore saturated upstream bandwidth through local dropping. Tunnels, egress, and dependencies such as DNS need explicit policies.

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Computer-security classifications

Historical TCSEC evaluates features/assurance under specified policies/configurations. TCB enforcement does not prove every business operation correct.

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Historical TCSEC assurance

The DoD Orange Book is not a universal modern product score. D fails higher classes. C1/C2 emphasize discretionary access/accountability; C2 adds individual control, auditing, and object reuse. B1–B3 add mandatory labels and stronger structure/analysis. A 1 adds formal design/verification and development assurance to B3 functionality.

B2 covert-channel requirements identify/analyze paths bypassing intended information flow, rather than support exfiltration. The TCB includes hardware, software, and firmware enforcing policy. If an outside component can break that policy, the boundary is wrong.

Evaluation applies to its version, configuration, policy, and environment. A 1 does not eliminate business errors/all attacks or automatically certify later versions. Formal proofs depend on their models, assumptions, and specifications.

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Windows 7 example

Separate tokens, ordered DACLs, MIC, and UAC. Remaining-mask checks and operation-specific integrity rules have different branches.

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Tokens and security descriptors

The historical primary token holds user/group SIDs and privileges. An impersonating server thread can use client context. Object descriptors contain owner, DACL, SACL; DACL grants/denies, while SACL can hold auditing/mandatory labels. Existing handles have granted access; not every operation reruns an identical open check.

For ordinary specific rights only, excluding owner/privilege exceptions, restricted-token checks, and object ACEs, set remaining=requested. In ACE order, matching deny intersecting remaining rejects. Matching allow removes granted bits. Empty remaining succeeds; reaching the end with bits left denies. This is not a universal union-all-allows-minus-all-denies algorithm.

DACL mask trace

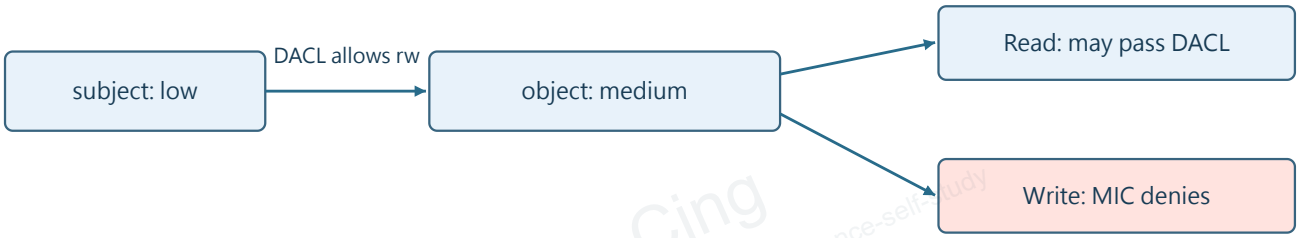
Request={r, w}; token U includes enabled Students. DACL=[deny(Students, w), allow(U, rw)]. Initially remaining={r, w}; the first intersects w and immediately denies. Without Students, skip it and allow U removes both bits, succeeding.

Deliberately placing allow(U, rw) first satisfies the request before reaching later deny. Deny does not override every allow regardless of order. Preferred canonical order places relevant explicit denies before allows; inheritance needs more ordering rules.

NULL DACL is absent and allows in this ordinary model. Empty DACL exists with no ACEs and grants nothing. Neither bypasses mandatory policy or special privilege rules.

MIC and UAC

One DACL grant; mandatory policy still distinguishes read/write



Assume NO_WRITE_UP only. Adding NO_READ_UP also denies the read branch.

Low-integrity subject, medium object, DACL allows r, w. With only NO_WRITE_UP, write fails while read can pass DACL/other checks. NO_READ_UP additionally denies read. Integrity primarily resists low-trust modification; it is not a full confidentiality-clearance model.

MIC and UAC (continued)

Windows 7 UAC can run an administrator's ordinary applications under a filtered medium token; controlled elevation obtains the corresponding higher token. A prompt neither creates rights the user never held nor identifies malicious programs.

Inheritance, copy/move, and propagation to existing objects depend on filesystem/API options. Inspect the final descriptor at creation. Parent ACL changes do not universally rewrite existing handles.

Composing security arguments

Authentication binds credentials/keys to subjects; authorization limits operations; cryptographic schemes protect messages; freshness/receiver state prevent replay; transaction/recovery connects authorization to a recoverable effect. One layer cannot supply another's missing premise.

Valid MAC replay still needs denial. TLS ending at a compromised proxy does not guarantee end-to-end secrecy. MIC can reject DACL-authorized writes. NX does not prevent authorization-data corruption. State premises, failure consequences, and observable checks.

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Proofs, examples, and configuration evidence

Fermat/CRT proves RSA correctness for all messages; a modular-power table checks one. Keystream leakage follows an XOR identity. Bayes connects IDS denominators; TPR alone is not precision. Historical cases retain versions, endpoints, and omissions. One replay test, DACL trace, or scan supports that observation without replacing general transition/threat-model reasoning.

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References

Silberschatz, Galvin, and Gagne, *Operating System Concepts*,9th edition, Chapter 15, printed pages657–702. Cases, diagrams, and exercises are original and contain no operational attack payloads.

RFC5246, § 7.4.7.1,7.4.9,8.1, F.1.1.2 covers historical TLS1.2 RSA premaster, Finished, and forward secrecy. RFC7568, § 3 prohibited SSL3 in 2015. Microsoft DACL checks and MIC document masks and integrity conditions. Windows 7 retains its historical scope without later AppContainer features.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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